

Java Collections Framework - Complete Cheat Sheet

Overview

The Java Collections Framework (JCF) provides a unified architecture for storing and manipulating groups of objects. It includes interfaces, implementations, and algorithms located in java.util package.

Core Interfaces

Core Interfaces:

- Collection (root interface)
- List (ordered, allows duplicates)
- Set (no duplicates)
- Queue (FIFO)
- Map (key-value pairs, not a subtype of Collection)

Implementations

Key Implementations:

- List: ArrayList, LinkedList, Vector, Stack
- Set: HashSet, LinkedHashSet, TreeSet
- Queue: PriorityQueue, ArrayDeque
- Map: HashMap, LinkedHashMap, TreeMap, Hashtable, ConcurrentHashMap

Internal Working & Memory

- ArrayList: Backed by dynamic array, default capacity 10, resizing factor 1.5x
- LinkedList: Doubly linked nodes (data + prev + next)
- HashMap: Array of buckets + linked list/tree for collisions, load factor 0.75
- TreeSet/TreeMap: Red-Black Tree (self-balancing)

Big-O Complexity

Structure	Access	Insert	Delete
ArrayList	O(1)	O(n)	O(n)
LinkedList	O(n)	O(1)	O(1)
HashSet	O(1)	O(1)	O(1)
TreeSet	O(log n)	O(log n)	O(log n)
HashMap	O(1)	O(1)	O(1)
TreeMap	O(log n)	O(log n)	O(log n)

Code Examples

ArrayList Example:

```
List list = new ArrayList<>();
```

```
list.add("Google");  
list.add("Amazon");  
System.out.println(list.get(0)); // Google
```

HashMap Example:

```
Map map = new HashMap<>();  
map.put("Apple", 1);  
map.put("Google", 2);  
System.out.println(map.get("Google")); // 2
```

Advanced Topics

- Fail-fast vs Fail-safe Iterators
- Concurrent Collections (ConcurrentHashMap, CopyOnWriteArrayList)
- Immutable Collections (Collections.unmodifiableList)
- Custom Comparator in TreeSet/TreeMap
- Memory Analysis (jmap, VisualVM)
- Thread Safety (Collections.synchronizedList(), ConcurrentHashMap)

Interview Questions

1. Difference between HashMap and ConcurrentHashMap?
2. How does HashMap handle collisions?
3. Why is ArrayList faster than LinkedList for random access?
4. Explain fail-fast iterator mechanism.
5. How does TreeMap maintain order?